

Implementation Fundamentals Accreditation Practical Lab

IMPORTANT: The **approximate build time for this lab is 15 minutes**. Please be patient. If the lab does not build after 30 minutes, try refreshing the page and rebuilding it again. If it does not build on the second attempt, contact ellearn@splunk.com.

Establish Deployment Client Connectivity

Maximum Estimated Duration: 8 hours

During this portion of the exercise, you will establish the forwarder instance as a deployment client of the all-in-one instance. The purpose of this in existing environments is to provide a single source of configuration data for what may be many thousands of forwarders. Manual maintenance or configuration of this many endpoints is impractical, so we employ configuration management tools. In this case, the tool available is Splunk's built-in Deployment Server. The all-in-one instance is already configured to act in this capacity, so the work that remains is to establish the forwarder's connection to that Deployment Server.

1. Download the "base_configs" from Google Drive using the following link:
<https://drive.google.com/drive/folders/107qWrfsv17j5bLxc21ymTagjtHG0AobF>
2. Customize the org_all_deployment app by first renaming it, replacing the "org" prefix with your initials.

During a customer engagement, you will utilize a customer-specific prefix to make it clear that the content was produced for that customer. For example, configurations created for Splunk, Inc. might utilize the stock ticker symbol "splk" as a prefix for these small config apps.

3. In the renamed deployment client app, adjust the targetUri parameter within the deploymentclient.conf contained in the app.
4. Now copy this modified deploymentclient app to the forwarder instance.

In order to keep the contents of the deploymentclient app (read: the address of the Deployment Server) up to date, we will later serve this same deploymentclient app from the Deployment Server itself. This would allow us to change the address in the future, if necessary.

5. Copy this same app to the deployment server itself.

NOTE: For the deployment server, apps awaiting deployment live within `/opt/splunk/etc/deployment-apps`

6. Now you will need to connect directly to the command line of your forwarder instance.

7. Validate that your configuration app has been copied into place correctly, by examining the contents of the `/opt/splunk/etc/apps` directory.

In addition to many default applications provided in the package, you should also see your `deploymentclient` app.

The `/opt/splunk` folder is owned by the 'splunk' user. Furthermore, the Splunk processes run as this user, so to ensure smooth operation, begin by changing the ownership of the apps you transferred.

8. `cd /opt/splunk/etc/apps && chown -R splunk`

To enable the deployment client functionality, the Splunk instance must start up with a `deploymentclient.conf` file in place. In order to trigger this behavior in your forwarder instance, you must restart Splunk for the changes there to take effect.

9. Within the shell, issue: `/opt/splunkforwarder/bin/splunk restart`

Check Your Work

1. Connect to the Splunk UI on the published port.
2. Log in to the Splunk GUI on the all-in-one.
3. Go to Settings > Forwarder Management and validate that the forwarder appears as a client. (It will be the only one.).

Identify Data Onboarding Parameters

Now that we can manage the configuration of the forwarder by utilizing the Forwarder Management GUI, prepare to onboard the data there by ascertaining the correct settings for this data type. Remember from your PS Best Practices eLearning modules, we always want to set the following six parameters in `props.conf`.

```
TIME_PREFIX
TIME_FORMAT
MAX_TIMESTAMP_LOOKAHEAD
LINE_BREAKER
SHOULD_LINEMERGE
TRUNCATE
```

Correct settings of these parameters are required for efficient and correct ingestion of events into Splunk. For data that is incorrectly ingested into Splunk, the only solution is to delete it and try again; these parameters cannot be adjusted once the data has been loaded!

Meet your data:

Target Index	SourceType	Data Directory	Number of Files
maildata	email_log	/var/log/mail	5

1. Connect to the running forwarder instance once more.
2. Navigate through the filesystem to the path `/var/log/mail/`
3. Examine the files contained here (for example, `maillog-smtp2.1.gz`) to figure out what the onboarding parameters should be.
4. You can view the contents of a gzipped file without unzipping it first by issuing a command like this:
`gzip -dc maillog-smtp2.1.gz | more`

NOTE: To help you identify the correct variables for you `TIME_FORMAT` setting Reference the “Enhanced `strptime()` Support” section of following two Splunk docs:
https://docs.splunk.com/Documentation/Splunk/latest/Data/Configuretimestamprecognition-Enhanced_strptime.28.29_support
<https://docs.splunk.com/Documentation/Splunk/latest/SearchReference/Commontimeformatvariables>

5. Clone the `org_APP_TEMPLATE` app for props settings.

NOTE: Make sure to call out the sourcetype of the data and props in the name of the configuration app.

6. Continue to figure out the appropriate parameters for onboarding this data source.
7. Place the `props.conf` you’ve worked up into the new app created in step 5.

NOTE: This app must be placed on the host that parses the data. In this kind of setup, the indexer (all-in-one) instance bears this responsibility.

Index-time properties (of which these data onboarding settings are part) require that the configurations of the instance be reloaded to take effect. In a production environment, this requires careful orchestration, but for the purposes of this test, simply restart Splunk on the all-in-one instance.

8. Connect to the all-in-one instance.
9. Restart Splunk:
`/opt/splunk/bin/splunk restart`

Define and Deploy Input Specification

The goal of this exercise is to create an appropriate `inputs.conf` configuration file to prompt the forwarder to ingest the provided data sample. Utilizing the base configs, you will create the necessary settings, and then employ the Deployment Server to place this configuration app on the forwarder instance.

All of the work in this module should take place on the all-in-one instance.

1. Create a new base config app for “maillog inputs” using the `org_APP_TEMPLATE`.

NOTE: Remember to use your initials as a prefix, and employ proper app naming.

You will create an `inputs.conf` stanza to monitor the directory containing the provided sample.

NOTE: Utilize the `org_dept_app_inputs` within the base configs you downloaded in the beginning of the lab for a sample `inputs.conf`.

2. Set the destination index to “maildata”.
3. Set the sourcetype of this data to “email_log”.
4. Copy this to the deployment-apps folder on the DS.

Now that we have an inputs specification, you must provide information to the forwarder on where to send the data it collects. This is a conf specification.

5. Clone the base config app “org_all_forwarder_outputs”, and adjust the ‘server’ parameter in the `tcpout:primary_indexers` group to reflect the (single) IP address of the all-in-one instance.
6. Copy this app into the deployment-apps folder on the DS as well.
7. Now, utilizing the Forwarder Management GUI, create a server class to contain the forwarder.

NOTE: While it’s true that this environment only has one forwarder, a production environment would have many, potentially thousands, and care should be taken to match only the hosts that are required.

8. For this exercise, include a whitelist statement to match your forwarder instance as the only entry to the class.
9. Add the two apps created above to the server class.
10. Note that changes to `conf` and/or `outputs.conf` require a restart of the node, so you should direct the app to “restart Splunkd” when it is deployed.
11. Additionally, keep the deployment client configuration up to date by assigning the customized “org_all_deploymentclient” app to the server class as well.
12. Once you apply your changes, the forwarder should restart, and begin reading in the mail data. Wait a few moments for this process to complete, then proceed with the next section: Create Search-Time Field Extractions.

Create Search-Time Extractions

The aim of this exercise is to extract meaningful fields from the provided data sample. As you've learned in your Splunk Education curriculum, Splunk can often interpret data automatically. However, depending upon the shape of the data, this may not be true, and it would then fall to an administrator (such as yourself) to craft the necessary configuration settings to extract the desired fields.

1. Search for the data you've ingested.

Observe that each event contains the string 'sm-mta' followed by a number enclosed in square brackets ([]), a colon, and followed in turn by a string of 14 characters and a colon.

2. Craft a field extraction to extract the digits as a new field called "pid".
3. Craft a field extraction to collect the 14-character string as a field called "qid".
4. Narrow the search for events by adding the keyword "milter" to the search bar.
5. Observe the returned events.

NOTE: Note that some feature the string "Milter: data, discard". An important metric for the administrators of this email system is the count of dropped or discarded messages.

6. Extract this "discard" string as a new field "milter_action".
7. Narrow these results even further by adding the additional keywords "add header" (your search string should read `index=maildata milter add header`).

NOTE: Some of these events feature a "probability" value indicating the likelihood that the message is, in fact, spam email.

8. Capture probability value (numeric part only) as a new field called "spam_pct".
9. These new settings also live in `props.conf` (and maybe `transforms.conf`, depending upon how you built the field extraction rules), so be sure to place these new entries in the appropriate app (the same one you created for the data onboarding props).

Build a Dashboard

One of Splunk's strong selling points is the ability to help users visualize data. Splunk dashboards provide a built-in mechanism to do just that. A wide array of graph types, tables, and other data representations are at your fingertips to help the customer get value from their machine data. In this exercise, you will build a dashboard utilizing best practices techniques learned in your eLearning to represent the data from this sample.

You will begin by creating a new app to contain your dashboard.

1. Open the Apps menu (top left), then select "Manage Apps".
2. On the apps page, click "Create app".
3. Call the app "email data" (with a folder name of email_data).
4. Build up search strings to match the following checklist of desired metrics.
 - 1a. Number (count only) of messages discarded
 - 1b. Number of messages delivered (stat=Sent) by hour.
 - 1c. Number of messages successfully delivered (count only).
 - 1d. A list of recipients of messages that where stat is not "Sent" showing the msgid, sender (from) and size.
 - 1e. The average sending delay, shown broken out by hour.
 - 1f. Top ten sending addresses by message count.
 - 1g. Top ten recipients by message count.
 - 1h. Top ten sending addresses by message size.
 - 1i. Top ten recipients by message size.

NOTE: It is not required to create saved searches / reports for these items. Instead, it is intended for you to prepare them for inclusion the dashboard.

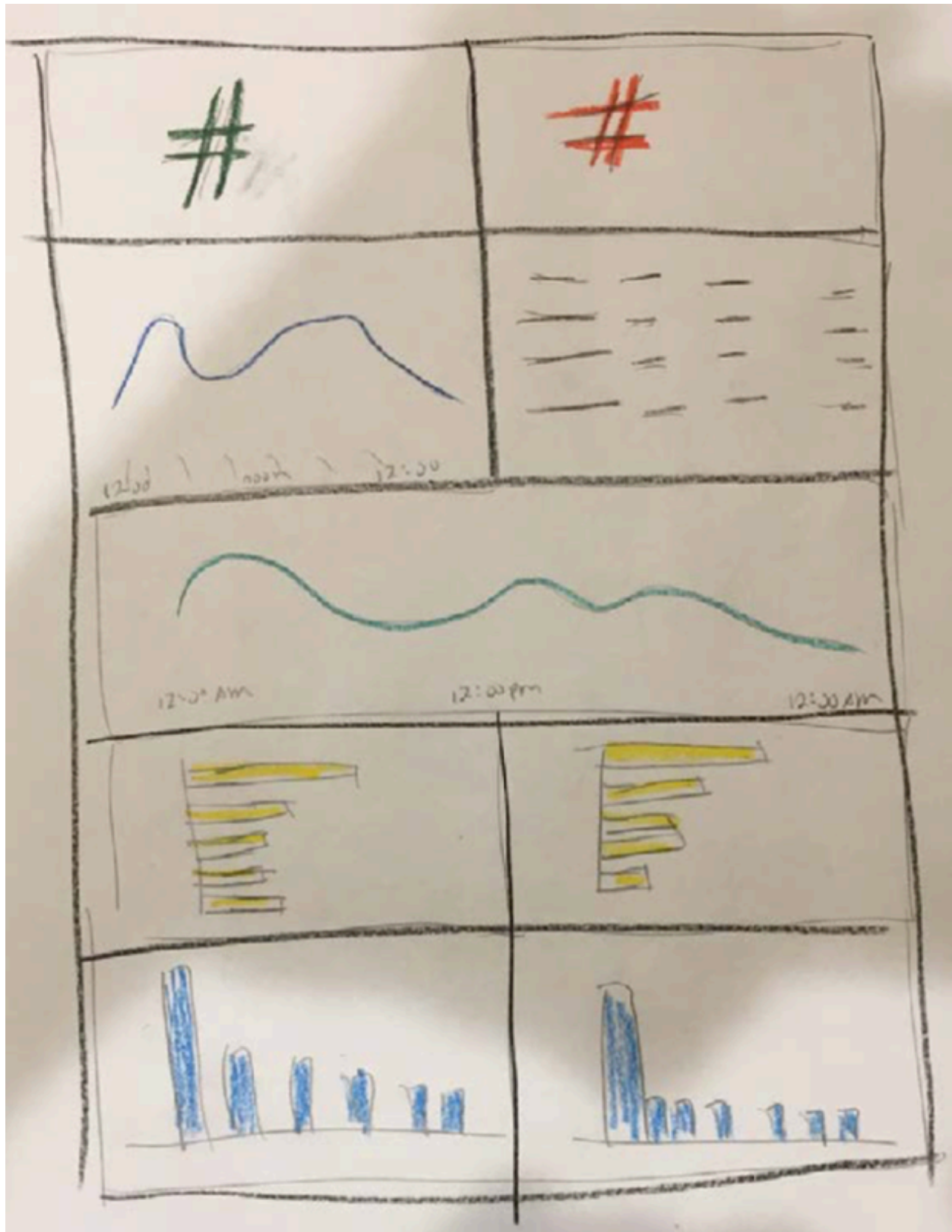
NOTE: Note that search 1i requires stitching together two distinct events (the first, containing the size, and the second, containing the recipient address) based upon a common queue ID ('qid' field). You will need to employ some means of linking the two together (transaction is easy, but expensive – stats is more efficient, but trickier) to calculate this result.

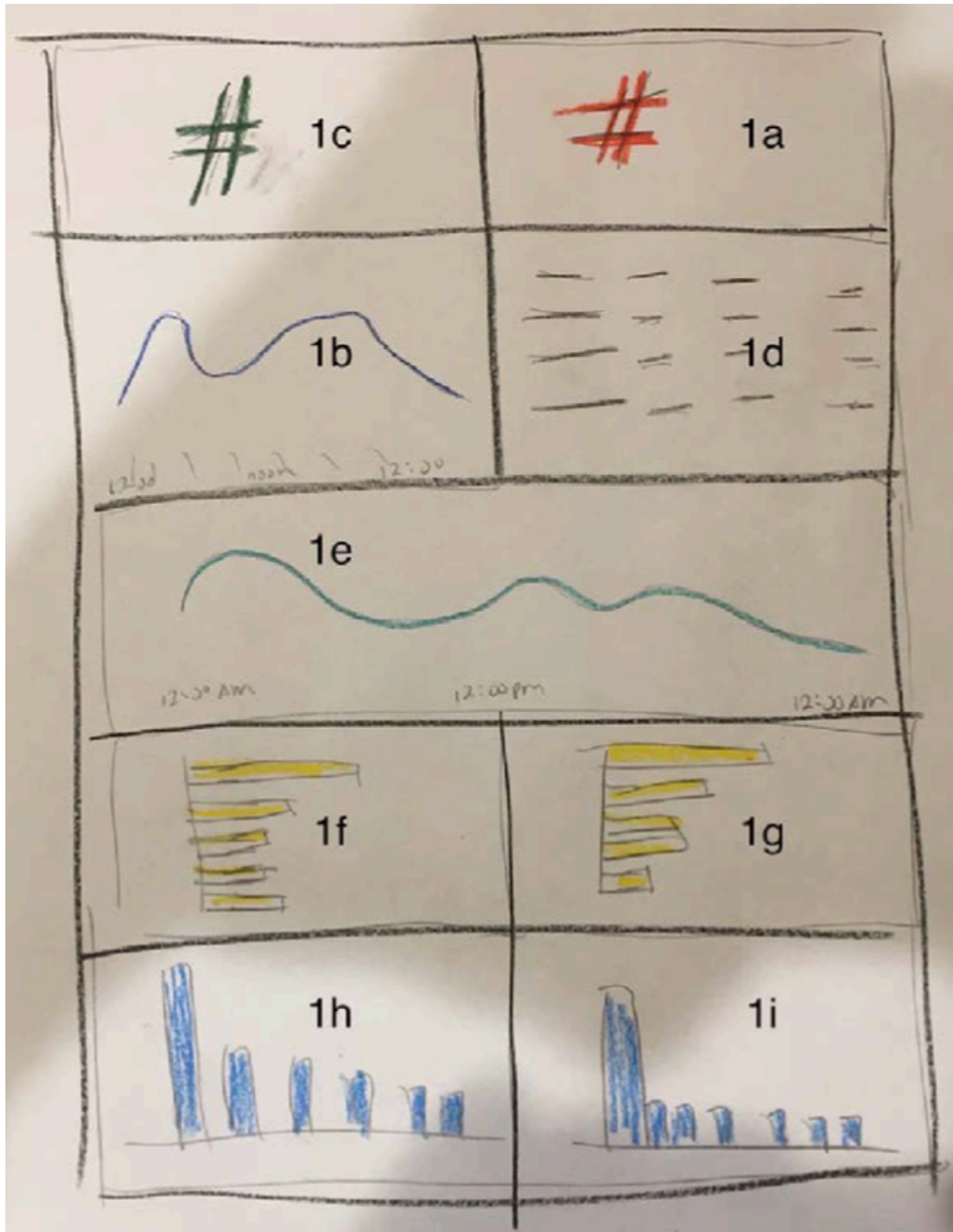
5. Build a dashboard employing the searches you authored in the previous step.
6. Save this dashboard with the name "mail_dashboard".
7. A sample workflow is shown below, starting with one of the searches above, click "Save As" in the top right of the search bar.
8. Then, select "Dashboard Panel". In the popup menu, this is where you specify the name of the dashboard.

The customer's specification is provided on a sketch, shown below. A second image is shown, marked up with the desired placement of each search. The markers in the second diagram are taken from the step shown above. For example the search string for "Number of messages delivered (stat=Sent) by hour" is noted as "1b".

Remember from your eLearning modules that a single dashboard, running many searches, can have an oversized impact on the system, despite being run by just one user.

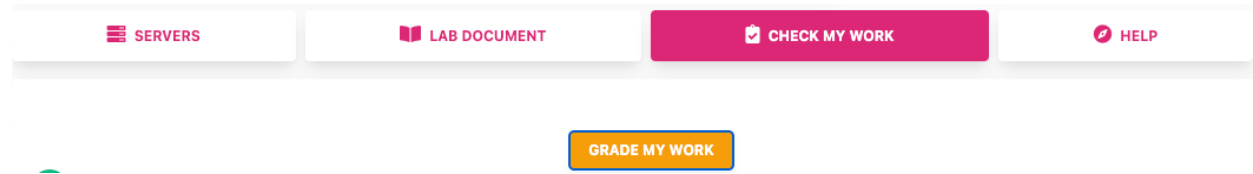
9. Think of ways to combine some of these searches (Is one a subset of another? Does one search just represent a "grand total" of another one?) and adjust your dashboard accordingly.





Grading Your Work

Grading of the lab is done through the same interface where your student instances are displayed.



Grading of the labs is an intensive process involving many calls to the instances. This may take several minutes to complete. Please be patient. If the grading does not complete in 5 minutes, please reach out to ellearn@splunk.com for assistance.

For best results, make sure you complete all of the lab steps before grading the lab.

1. Click the Check My Work Tab.
2. Click the Grade My Work button.

After a few minutes, your results will be displayed.

You must pass all graded items to receive credit for the lab. Resolve any failed grading items and re-issue the grading check.

If you pass all of the lab grading checks, the Learning Management System (LMS) will terminate the browser window. Log back into the LMS to view your completed status for the lab.